

Testing and Continuous Integration Lab Report

Glen Testamark

[Fork Repository](#)

[Upstream Repository](#)

Phase 1: Test Coverage

The screenshot displays a VS Code editor with the Explorer sidebar on the left showing the project structure. The main editor window shows the file `test_account.py` with the following Python code:

```
1 """
2 Test Cases for Account Model
3 """
4 import json
5 from pathlib import Path
6 import pytest
7 from models import db
8 from models.account import Account, DataValidationError
9
10 ACCOUNT_DATA = {}
11
12 @pytest.fixture(scope="module", autouse=True)
13 def load_account_data():
14     """ Load data needed by tests """
15     global ACCOUNT_DATA
16     with open(Path(__file__).parent / 'fixtures' / 'account_data.json') as json_data:
17         ACCOUNT_DATA = json.load(json_data)
18
19     # Set up the database tables
20     db.create_all()
21     yield
22     db.session.close()
23
24 @pytest.fixture
25 def setup_account():
26     """Fixture to create a test account"""
27     account = Account(name="John Businge", email="john.businge@example.com")
28     db.session.add(account)
29     db.session.commit()
30     return account
31
```

The bottom panel shows the terminal output of a successful pytest run with coverage:

```
> python -m pytest --cov=models --cov-report=term-missing
===== test session starts =====
platform linux -- Python 3.14.7, pytest-8.4.2, pluggy-1.6.0 -- /home/dev/class/venv/bin/python
cachedir: .pytest_cache
rootdir: /home/dev/class/Lab1/test_coverage_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 5 items

tests/test_account.py::test_account_role_assignment PASSED [ 20%]
tests/test_account.py::test_invalid_role_assignment PASSED [ 40%]
tests/test_account.py::test_to_dict PASSED [ 60%]
tests/test_account.py::test_required_fields PASSED [ 80%]
tests/test_account.py::test_password_hashing PASSED [100%]

===== tests coverage =====
----- coverage: platform linux, python 3.14.7-final-0 -----
Name                               Stmts  Miss  Cover   Missing
-----
models/__init__.py                   7      0  100%
models/account.py                   58     19   67%   30, 47-49, 60-62, 66-68, 72-76, 94, 98, 102-103
tests/__init__.py                    0      0  100%
tests/test_account.py               57      4   93%   27-30
-----
TOTAL                               122     23   81%

===== 5 passed in 0.35s =====
~/c/Lab1/test_coverage_lab test_coverage_lab_geeklay *4 ?1 >
```

Screenshot showing the `test_coverage_lab` folder and a successful run of pytest.

- Setup PR issue: <https://github.com/kenner1-unlv/Group-2/issues/43>

- Setup PR: <https://github.com/kenner1-unlv/Group-2/pull/44>
- Links to my issues
 - <https://github.com/kenner1-unlv/Group-2/issues/56>
 - <https://github.com/kenner1-unlv/Group-2/issues/58>
- Link to PR & results: <https://github.com/kenner1-unlv/Group-2/pull/73>

Brief Description of Test and Behavior it Verifies

test_to_dict() tests the serialization of an **Account** into a dictionary. First it creates an **Account** and ensures that it is a dict, checks if the expected (required) keys are in the dictionary and whether or not the values for name and email match the original **Account** object.

test_required_fields() tests ***validate_required_fields()*** by making sure that it throws a `ValidationError` when the email or name fields are uninitialized (have default values).

- Teammates PR & Review Comment:
<https://github.com/kenner1-unlv/Group-2/pull/48>
<https://github.com/kenner1-unlv/Group-2/pull/48#pullrequestreview-5203494838>

Feedback and Addressing Feedback

I received a review from Sokrat who gave good feedback regarding the verification of serialization for ***to_dict()***. Taking that feedback into account I changed the test to account for verifying the proper serialization of the Account. After doing so it was stated that the changes met the requirements.

- Approved PR or merge commit:
<https://github.com/kenner1-unlv/Group-2/commit/2e7bbe5665aee6d6d90b9770926278ec81bca144>

Phase 2: Test Driven Development

The screenshot shows the VS Code interface with the Explorer view on the left displaying the project structure. The main editor shows the `counter.py` file with the following code:

```
11 def counter_exists(name):
12     return name in COUNTERS
13
14
15 @app.route('/counters/<name>', methods=['POST'])
16 def create_counter(name):
17     """Create a counter"""
18     if counter_exists(name):
19         return jsonify({"error": f"Counter {name} already exists"}), status.HTTP_409_CONFLICT
20     COUNTERS[name] = 0
21     return jsonify({"name": COUNTERS[name]}), status.HTTP_201_CREATED
22
23 @app.route('/counters/<name>', methods=['DELETE'])
24 def delete_counter(name: str):
25     """Delete a counter from COUNTERS"""
26     return jsonify({"Success": f"Counter {name}: {COUNTERS.pop(name)} deleted"}), status.HTTP_
27
28
29 @app.route('/counters', methods=['GET'])
30 def list_counters():
31     """List all counters"""
32     return jsonify(COUNTERS), status.HTTP_200_OK
33
```

The terminal view at the bottom shows the output of a `pytest` command:

```
cd ..
cd tdd_lab
pytest
===== test session starts =====
platform linux -- Python 3.14.7, pytest-8.4.2, pluggy-1.6.0 -- /home/dev/class/venv/bin/python
cachedir: .pytest_cache
rootdir: /home/dev/class/Lab1/tdd_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 3 items

tests/test_counter.py::TestCounterEndpoints::test_create_counter PASSED [ 33%]
tests/test_counter.py::TestCounterEndpoints::test_delete_counter PASSED [ 66%]
tests/test_counter.py::TestCounterEndpoints::test_list_counters PASSED [100%]

===== tests coverage =====
coverage: platform linux, python 3.14.7-final-0

Name          Stmts  Miss  Cover   Missing
-----
src/__init__.py      3      0  100%
src/counter.py     18      1   94%    19
src/status.py        6      0  100%
TOTAL              27      1   96%

===== 3 passed in 0.09s =====
```

Screenshot showing the tdd_lab folder and a run of pytest

- Setup PR issue: <https://github.com/kenner1-unlv/Group-2/issues/46>
- Setup PR: <https://github.com/kenner1-unlv/Group-2/pull/47>
- Link to my issue: <https://github.com/kenner1-unlv/Group-2/issues/63>
- Link to PR: <https://github.com/kenner1-unlv/Group-2/pull/74>

The feature being tested is deletion of a counter – **DELETE /counters/<name>**. The delete feature adds an endpoint, **DELETE /counters/<name>** that deletes a counter and returns two responses: a successful deletion JSON response, “[...] {name}: {counter count},” and **HTTP_204_NO_CONTENT**.

- **RED, GREEN, REFACTOR Evidence:**

- <https://github.com/kenner1-unlv/Group-2/pull/74#issue-5470137677>
- Refactor comments:

There was actually no need to refactor, since this specific delete feature item does not share any code from other feature items. However, bundling the deletion (pop) into the format string, since the decision was made to include the counter value in the json response, reduced the function to a single line.

Feedback and Addressing Feedback

I received a review from Sokrat, who verified the changes locally and stated that it met the requirements. I initially forgot to link to the opened issue and Sokrat added it for me. Once I read the comment, I edited the PR to include the issue as a measure of redundancy and completeness.

- Teammates PR and Review Comment:
 - <https://github.com/kenner1-unlv/Group-2/pull/50>
 - <https://github.com/kenner1-unlv/Group-2/pull/50#pullrequestreview-5203787592>

The evidence examined while reviewing included the RED phase correctly failing and corresponding with the PR showing ‘**E assert 404 == 200.**’ The GREEN phase results matched those of the PRs local GREEN phase run.

- Approved PR or merge commit:
<https://github.com/kenner1-unlv/Group-2/commit/e34f2740c62b8787232040aaf1e24db4b5ceebe9>

Phase 3: CI

- Link to fork: https://github.com/geeklay/Lab1/tree/ci_lab1
- Evidence of upstream synchronization (unfortunately did not screenshot merging)

```
commit 960d713fdc5a2a2f14cdefc8fef1c3c2ba682dc7 (HEAD -> ci_lab1, origin/ci_lab1)
Author: geeklay <13053807+geeklay@users.noreply.github.com>
Date:   Wed Sep 16 22:41:12 2026 -0700

    update after merge #1

commit 62aeb2cc0412b0c250b35568dd4f5680d4ea0caa
Merge: 5696521 e34f274
Author: geeklay <13053807+geeklay@users.noreply.github.com>
Date:   Wed Sep 16 22:29:51 2026 -0700

    merge and resolve conflicts

commit e34f2740c62b8787232040aaf1e24db4b5ceebe9 (upstream/main, upstream/HEAD)
Merge: e05079c 6b4a828
Author: rosto-designs <sokratrostopmyan@gmail.com>
Date:   Wed Sep 16 22:00:16 2026 -0700

    Merge pull request #74 from geeklay/tdd

    Add DELETE /counter/<name> endpoint to delete a counter

commit e05079cc16d4583573d92aebc51cc4a7dea871f2
Merge: 3bc668d 4c4a0d4
Author: kenner1-unlv <kenner1@unlv.nevada.edu>
Date:   Wed Sep 16 21:46:32 2026 -0700

    Merge pull request #55 from MikePodo/phase3-ci-matrix-mikepodo

    CI Enhancement 1: Run tests across a Python 3.9, 3.10, and 3.11 matrix

commit 3bc668dc09c840abd9710483a7900fc076fd63f1
Merge: 2e7bbe5 b571011
Author: kenner1-unlv <kenner1@unlv.nevada.edu>
Date:   Wed Sep 16 21:44:43 2026 -0700

    Merge pull request #62 from kenner1-unlv/phase2-reset-counters-kenner1-unlv

    Add endpoint to reset all counters

commit 2e7bbe5665aee6d6d90b9770926278ec81bca144
Merge: 14e7326 b8019b5
Author: rosto-designs <sokratrostopmyan@gmail.com>
Date:   Wed Sep 16 21:39:51 2026 -0700

    :
```

- Evidence of ci_lab starter workflow running successfully:

```
> pytest
===== test session starts =====
platform linux -- Python 3.14.7, pytest-8.4.2, pluggy-1.6.0 -- /home/dev/class/venv/bin/python
cachedir: .pytest_cache
rootdir: /home/dev/class/Lab1/ci_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 22 items

tests/test_counter.py::TestCounterEndpoints::test_create_counter PASSED [ 4%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_duplicate_counter PASSED [ 9%]
tests/test_counter.py::TestCounterEndpoints::test_retrieve_existing_counter PASSED [ 13%]
tests/test_counter.py::TestCounterEndpoints::test_return_404_for_non_existent_counter PASSED [ 18%]
tests/test_counter.py::TestCounterEndpoints::test_increment_counter PASSED [ 22%]
tests/test_counter.py::TestCounterEndpoints::test_prevent Updating non-existent counter PASSED [ 27%]
tests/test_counter.py::TestCounterEndpoints::test_delete_counter PASSED [ 31%]
tests/test_counter.py::TestCounterEndpoints::test_prevent deleting non-existent counter PASSED [ 36%]
tests/test_counter.py::TestCounterEndpoints::test_reset_all_counters PASSED [ 40%]
tests/test_counter.py::TestCounterEndpoints::test_list_all_counters PASSED [ 45%]
tests/test_counter.py::TestCounterEndpoints::test_handle_invalid_http_methods PASSED [ 50%]
tests/test_counter.py::TestCounterEndpoints::test_get_total_counters PASSED [ 54%]
tests/test_counter.py::TestCounterEndpoints::test_top_n_counters PASSED [ 59%]
tests/test_counter.py::TestCounterEndpoints::test_bottom_n_counters PASSED [ 63%]
tests/test_counter.py::TestCounterEndpoints::test_set_counter_to_value PASSED [ 68%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_negative_counter_values PASSED [ 72%]
tests/test_counter.py::TestCounterEndpoints::test_reset_single_counter PASSED [ 77%]
tests/test_counter.py::TestCounterEndpoints::test_prevent resetting non-existent counter PASSED [ 81%]
tests/test_counter.py::TestCounterEndpoints::test_get_total_number_of_counters PASSED [ 86%]
tests/test_counter.py::TestCounterEndpoints::test_counters_greater_than_threshold PASSED [ 90%]
tests/test_counter.py::TestCounterEndpoints::test_counters_less_than_threshold PASSED [ 95%]
tests/test_counter.py::TestCounterEndpoints::test_validate_counter_name PASSED [100%]

===== tests coverage =====
coverage: platform linux, python 3.14.7-final-0

Name                Stmts   Miss  Cover   Missing
src/__init__.py         3      0  100%
src/counter.py        87      5   94%    71, 85, 96, 99-100
src/status.py          7      0  100%

TOTAL                  97      5   95%

22 passed in 0.13s

~/class/Lab1/ci_lab on @a331fb1b #4 ?1 > |
```

- Link to my issue: <https://github.com/kenner1-unlv/Group-2/issues/84>
- Link to PR: <https://github.com/kenner1-unlv/Group-2/pull/87>
- Link to workflow excerpt:
https://github.com/geeklay/Lab1/blob/ci_lab1/.github/workflows/ci.yml
- Link to failing CI log:
 - [Fork Before Merge Conflict](#) [Fork After Merge Conflict Resolution](#)
 - Did not PR a controlled failure to group repository before fixing on fork so PR would show a failure in groups repository Actions.
 - The job failed because coverage.json was not generated during a pytest run, since the flag '--cov-report=json' was missing.
- Link to fix commit:
<https://github.com/kenner1-unlv/Group-2/commit/5696521f2c6385f734606f52fafb8cda8613d9ad>
- Link to successful CI run:
<https://github.com/kenner1-unlv/Group-2/actions/runs/35186810353>
- Teammates PR and Comments:
 - <https://github.com/kenner1-unlv/Group-2/pull/55>
 - <https://github.com/kenner1-unlv/Group-2/pull/55#pullrequestreview-5229305834>

I reviewed the PR by first checking the commit diffs. The workflow was changed appropriately to include the matrix of python versions: 3.9, 3.10, and 3.1. The change also ensured that the tests would not hamper the others by setting fail-fast to false. I then verified that the tests ran locally, but failed in CI by checking the failed workflow runs in Actions.

Reflection

The lab taught me that testing is extremely useful for determining if a functionality works as intended. Furthermore, a test that passes does not necessarily test the intended functionality, as evidenced by my first PR. The tdd portion taught me that it a RED and GREEN phase are highly useful when doing test driven development. The CI portion taught me about CI in general and how it can catch issues that you otherwise wouldn't be able to. Also, merge conflicts are a massive pain and it seems as though they should be avoided as much as possible.